

eleVADR Detection Module Reference

Compact technical guide to the 75 detection modules, their inputs, and how Zeek evidence is used.

Important: Zeek is the network telemetry engine; it generally does *not* issue the eleVADR finding itself. Zeek protocol analyzers turn packets into structured logs. eleVADR normalizes those logs into an AnalysisContext, combines them with the Detection Context (assets, segments, trusted infrastructure, authorization and allowlists), then each detector applies its rule. Observed Zeek traffic is evidence only - it does not create authorization or policy.

Common Zeek evidence

Log(s)	Typical fields used
conn.log	Endpoints, ports/service, timestamps, bytes/packets, connection state, VLANs
dns/http/ssl	Application queries/requests; TLS SNI, versions, ciphers and fingerprints
x509/files	Certificate identity/validity/key details; extracted file metadata and endpoints
modbus/dnp3/enip/s7comm	Industrial protocol operations, function/service codes and errors
arp/dhcp/ntp	L2 identity, address assignment, and time-source behavior
weird + auth logs	Protocol violations plus SSH/Kerberos/NTLM/RDP/VNC authentication/session evidence

Reading each entry

Inputs lists the Zeek logs and the most relevant Detection Context/policy data. **Zeek evidence** summarizes the packet-derived fields made available to the detector. **Detection** states what eleVADR evaluates using that evidence.

Detection modules 1-8

01. ARP / IP-MAC Identity Change

arp_ip_mac_identity_change

Inputs: Zeek: arp.log. Context/policy: asset inventory, arp identity policy.

Zeek evidence: **arp.log** and supplies sender/target IP-MAC pairs, opcode and timestamp.

Detection: Detects ARP observations where an authoritative asset IP is claimed by an unexpected MAC address, or where one IP rapidly appears behind multiple MAC addresses.

03. BACnet Who-Is/I-Am Flood or External Discovery

bacnet_discovery_anomalies

Inputs: Zeek: bacnet.log. Context/policy: segment/zone definitions, bacnet discovery policy.

Zeek evidence: **bacnet.log** and supplies Who-Is/I-Am and related discovery source/destination activity.

Detection: Detects abnormal BACnet Who-Is/I-Am discovery bursts and BACnet discovery activity originating from explicitly non-BAS network segments.

05. Brute Force Authentication

brute_force_authentication

Inputs: Zeek: ssh.log, rdp.log, http.log, kerberos.log, ntlm.log, vnc.log, conn.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **ssh.log** (SSH authentication/session evidence); **rdp.log** (RDP negotiation and NLA/CredSSP indicators); **http.log** (method, host/URI, User-Agent, status, body length and flow UID); plus kerberos.log, ntlm.log, vnc.log, conn.log.

Detection: Identifies repeated authentication failures against SSH, RDP, HTTP, Kerberos, SMB/NTLM, and VNC that are consistent with automated password guessing or credential spraying.

07. CoDeSys Runtime Exposure

codesys_runtime_exposure

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, allowed host pairs, codesys runtime policy, allowed/ignored hosts.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies CODESYS programming/runtime and gateway communications using vendor-documented default ports or explicit service labels, with higher severity when runtime access reaches configured OT controllers from public or non-OT sources.

02. ARP / L2 Reconnaissance

arp_l2_reconnaissance

Inputs: Zeek: arp.log. Context/policy: arp reconnaissance policy.

Zeek evidence: **arp.log** and supplies sender/target IP-MAC pairs, opcode and timestamp.

Detection: Detects hosts rapidly probing many distinct IPv4 targets with ARP requests.

04. Beaconing / C2 Communication

beaconing_c2_communication

Inputs: Zeek: conn.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies repeated outbound connections to the same destination at consistent intervals, a behavioral pattern commonly associated with malware command-and-control check-ins.

06. Cleartext Credentials

cleartext_credentials

Inputs: Zeek: ftp.log, http.log, smtp.log, telnet.log, login.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **ftp.log** (FTP commands, arguments and transfer endpoints); **http.log** (method, host/URI, User-Agent, status, body length and flow UID); **smtp.log** (SMTP commands/authentication metadata); plus telnet.log, login.log.

Detection: Identifies cleartext credential exposure in FTP, HTTP, SMTP, and Telnet analysis data. Password values are redacted from results.

08. Control System to Enterprise IT Communication (Non-DMZ)

control_system_enterprise_non_dmz

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, asset inventory, allowed host pairs, allowed segment pairs.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies direct communication between configured control-system/OT assets and enterprise or other non-OT networks when neither observed endpoint is an approved control-system DMZ intermediary.

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09. Controller Communication Jitter

controller_communication_jitter

Inputs: Zeek: conn.log. Context/policy: asset inventory, controller hosts, controller communication jitter policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects increased timing variance in established controller communications after a baseline.

10. Cross-Purdue-Level Traffic

cross_purdue_level_traffic

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, Purdue-level policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies direct communications between Purdue levels that should not communicate directly, including control levels reaching enterprise levels without an intermediate industrial DMZ.

11. Database Service Exposed

database_service_exposed

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, allowed host pairs, allowed segment pairs, database exposure policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies database services reachable across configured network-segment boundaries or from public Internet addresses, including MySQL, SQL Server, PostgreSQL, MongoDB, Redis, Memcached, and Elasticsearch.

12. Deprecated / Insecure Connection Services and Protocols

deprecated_insecure_services_protocols

Inputs: Zeek: smtp.log, ftp.log, http.log, telnet.log, login.log, conn.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **smtp.log** (SMTP commands/authentication metadata); **ftp.log** (FTP commands, arguments and transfer endpoints); **http.log** (method, host/URI, User-Agent, status, body length and flow UID); plus telnet.log, login.log, conn.log.

Detection: Identifies observed use of legacy, plaintext, or otherwise insecure connection services and insecure connection protocols.

13. PPTP / Deprecated VPN Protocol

deprecated_vpn_protocol

Inputs: Zeek: conn.log. Context/policy: allowed host pairs, deprecated vpn policy, allowed/ignored hosts.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies PPTP using TCP/1723 with GRE when available, and L2TP sessions on UDP/1701 that lack correlated IKE/IPsec evidence.

14. DNS Resolver Source Drift

dns_source_drift

Inputs: Zeek: dns.log. Context/policy: trusted DNS resolvers, dns source drift policy.

Zeek evidence: **dns.log** and supplies client/resolver, query name/type, reply code and timestamp.

Detection: Detects OT clients that use DNS resolvers outside explicitly trusted infrastructure or change resolver after a capture baseline.

15. DNS Tunneling / Exfiltration

dns_tunneling_exfiltration

Inputs: Zeek: dns.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **dns.log** and supplies client/resolver, query name/type, reply code and timestamp.

Detection: Identifies DNS behavior consistent with covert tunneling or exfiltration, including encoded/high-entropy subdomains, high-payload TXT/NULL/ANY queries, and sustained NXDOMAIN-heavy query patterns.

16. Encrypted Session Fingerprint Change

encrypted_session_fingerprint_change

Inputs: Zeek: ssl.log. Context/policy: encrypted session fingerprint policy, ignored hosts.

Zeek evidence: **ssl.log** and supplies TLS endpoints, SNI, version/cipher, fingerprint/JA3 data and certificate references.

Detection: Detects established TLS endpoint pairs whose observed TLS fingerprint changes after a capture baseline.

Detection modules 17-24

17. Engineers Tools Over Clear-Text (TFTP/FTP)

engineering_tools_cleartext

Inputs: Zeek: ftp.log, tftp.log, files.log. Context/policy: engineering cleartext policy, ftp tftp policy.

Zeek evidence: **ftp.log** (FTP commands, arguments and transfer endpoints); **tftp.log** (operation, filename and endpoints); **files.log** (file UID, name/MIME, size and transmitter/receiver hosts).

Detection: Detects firmware or configuration material transferred using clear-text FTP or TFTP, using Zeek FTP/TFTP/file metadata rather than port-only inference.

19. EtherNet/IP CIP Write and Session Abuses

enip_cip_write_session_abuses

Inputs: Zeek: enip.log. Context/policy: authorized write paths, enip cip policy, approved CIP service codes.

Zeek evidence: **enip.log** and supplies originator/target, CIP service/status and Forward_Open activity.

Detection: Detects explicit CIP write services and excessive Forward_Open/Large_Forward_Open activity in EtherNet/IP telemetry, with optional policy checks for approved write paths.

21. File Extraction on Wire (Sensitive Types)

file_extraction_sensitive_types

Inputs: Zeek: files.log. Context/policy: segment/zone definitions, approved external destinations, file transfer policy, sensitive file policy.

Zeek evidence: **files.log** and supplies file UID, name/MIME, size and transmitter/receiver hosts.

Detection: Detects executable, configuration, or archive file types observed by Zeek files.log when they are transferred to external destinations or across explicitly classified OT/IT boundaries.

23. HTTP User-Agent Anomalies

http_user_agent_anomalies

Inputs: Zeek: http.log. Context/policy: segment/zone definitions, http user agent policy.

Zeek evidence: **http.log** and supplies method, host/URI, User-Agent, status, body length and flow UID.

Detection: Identifies CLI/scripted HTTP clients and rare or missing User-Agent strings on sensitive HTTP flows using protocol telemetry and conservative baselines.

18. Engineering Workstation Control Burst

engineering_workstation_control_burst

Inputs: Zeek: modbus.log, dnp3.log, s7comm.log, enip.log. Context/policy: asset inventory, authorized control paths, engineering workstations, engineering control burst policy.

Zeek evidence: **modbus.log** (client/server, function/operation, exception/status and time); **dnp3.log** (master/outstation, function/control action and IIN/error flags); **s7comm.log** (client/PLC, function/operation and S7 error/result); plus enip.log.

Detection: Detects bursts of control-write/programming operations from authoritative engineering workstations, including bursts on otherwise authorized paths.

20. Excessive Broadcast / Multicast in OT

excessive_broadcast_multicast_ot

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, broadcast multicast ot policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects unusually high broadcast or multicast traffic rates from explicitly configured OT segments/VLANs where point-to-point traffic is expected to dominate. Uses Zeek conn.log packet counts when available and falls back to flow-event rates when packet counts are unavailable.

22. High Fan-Out or Fan-In Patterns

high_fan_in_out

Inputs: Zeek: conn.log. Context/policy: fan in out policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies bursts where one host contacts many distinct destinations (fan-out), or many distinct sources suddenly contact one newly observed destination (fan-in).

24. ICCP / Inter-Control-Center Protocol

iccp_tase2_detected

Inputs: Zeek: mms.log, conn.log. Context/policy: allowed host pairs, iccp tase2 policy, allowed/ignored hosts.

Zeek evidence: **mms.log** (MMS/ISO-on-TCP service evidence); **conn.log** (endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags).

Detection: Identifies traffic consistent with ICCP/TASE.2, used for real-time data exchange between electric utility SCADA control centers. Explicit ICCP/TASE.2 service labels are preferred; TCP/102/OSI-on-TCP is treated conservatively because other industrial protocols also use ISO-on-TCP.

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25. ICMP as Data Channel

icmp_data_channel

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, icmp data channel policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies sustained ICMPv4 echo traffic with unusually uniform packet sizes and regular timing to external or explicitly unexpected destinations.

26. ICS Protocol Error / Exception Spike

ics_protocol_error_spike

Inputs: Zeek: modbus.log, dnp3.log, enip.log, s7comm.log. Context/policy: allowed host pairs, allowed/ignored hosts.

Zeek evidence: **modbus.log** (client/server, function/operation, exception/status and time); **dnp3.log** (master/outstation, function/control action and IIN/error flags); **enip.log** (originator/target, CIP service/status and Forward_Open activity); plus s7comm.log.

Detection: Detects elevated rates of ICS protocol error responses, including Modbus exceptions, DNP3 IIN error indications, EtherNet/IP/CIP general-status failures, and S7 error returns.

27. Modbus/DNP3 Write Operations Outside Allowed Paths

ics_write_operations

Inputs: Zeek: modbus.log, dnp3.log. Context/policy: authorized paths, ics write policy, approved function codes.

Zeek evidence: **modbus.log** (client/server, function/operation, exception/status and time); **dnp3.log** (master/outstation, function/control action and IIN/error flags).

Detection: Detects Modbus write operations and DNP3 write/control operations outside configured source-to-destination paths, plus protocol function codes outside an optional approved profile.

28. Internet-Exposed ICS Device / External Admin to PLC/HMI

internet_exposed_ics

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, asset inventory, allowed host pairs, approved external destinations.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies configured ICS assets communicating directly with public IP addresses and Internet-originated administrative access to PLC, RTU, HMI, BAS, historian, or other OT assets.

29. IPv6 Traffic in OT Environment

ipv6_traffic_ot

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, IPv4-only expectation, ipv6 ot policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies IPv6 communications observed in an OT/ICS environment that is explicitly documented as IPv4-only, including link-local, deprecated site-local, unique-local, multicast, and global IPv6.

30. IRC Traffic Detected

irc_traffic_detected

Inputs: Zeek: conn.log. Context/policy: allowed host pairs, irc traffic policy, allowed/ignored hosts.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies Internet Relay Chat traffic on standard IRC ports or on non-standard ports when Zeek explicitly identifies the service as IRC.

31. JA3/JA3S Fingerprint Outliers

ja3_fingerprint_outliers

Inputs: Zeek: ssl.log. Context/policy: segment/zone definitions, ja3 outlier policy, tls fingerprint policy.

Zeek evidence: **ssl.log** and supplies TLS endpoints, SNI, version/cipher, fingerprint/JA3 data and certificate references.

Detection: Detects rare TLS client/server JA3 or JA3S fingerprints that fall outside an established, stable fingerprint baseline for a configured network segment.

32. Kerberos AS-REP Roastable Accounts

kerberos_asrep_roastable_accounts

Inputs: Zeek: kerberos.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **kerberos.log** and supplies principal, message/error and pre-authentication evidence.

Detection: Detects Kerberos accounts explicitly observed allowing AS authentication without pre-authentication, a condition associated with AS-REP roasting.

Detection modules 33-40

33. Large Outbound POST/PUT to Unknown Hosts

large_outbound_http_uploads

Inputs: Zeek: http.log. Context/policy: segment/zone definitions, http upload policy, large http upload policy.

Zeek evidence: **http.log** and supplies method, host/URI, User-Agent, status, body length and flow UID.

Detection: Identifies substantial HTTP POST/PUT uploads from internal clients to external, non-approved destinations using explicit request-body length when available and conn.log orig_bytes only as a conservative fallback.

35. LLMNR/NBT-NS/mDNS Poisoning Signals

llmnr_nbns_mdns_poisoning_signals

Inputs: Zeek: dns.log. Context/policy: segment/zone definitions, name resolution poisoning policy.

Zeek evidence: **dns.log** and supplies client/resolver, query name/type, reply code and timestamp.

Detection: Detects unusually frequent multicast/broadcast name-resolution queries and suspicious LLMNR, NBT-NS, or mDNS responder patterns that can be consistent with name-resolution poisoning.

37. New OT Conversation Pair (Comm-Matrix Drift)

new_ot_conversation_pair

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, asset inventory, allowed host pairs, explicit OT hosts.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Builds a directed source-to-destination communication matrix during an OT baseline period and identifies new talker relationships that first appear after the baseline.

39. Niagara Fox Protocol Detected

niagara_fox_detected

Inputs: Zeek: ssl.log, conn.log. Context/policy: allowed host pairs, niagara fox policy, allowed/ignored hosts.

Zeek evidence: **ssl.log** (TLS endpoints, SNI, version/cipher, fingerprint/JA3 data and certificate references); **conn.log** (endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags).

Detection: Identifies Tridium Niagara Fox communications on TCP/1911 and Fox-over-SSL on TCP/4911, commonly used between JACE controllers and supervisory stations in building automation systems.

34. LDAP Cleartext / Anonymous Bind / Cross-Segment

ldap_cleartext_anonymous_cross_segment

Inputs: Zeek: ldap.log, ldap_bind.log, ldap_search.log, conn.log. Context/policy: segment/zone definitions.

Zeek evidence: **ldap.log** (LDAP request/session data); **ldap_bind.log** (bind mechanism/result); **ldap_search.log** (directory query metadata); plus conn.log.

Detection: Identifies LDAP anonymous or simple binds without TLS, cleartext LDAP on port 389, and LDAP communications that cross configured network segments, including OT-to-enterprise directory queries.

36. NetBIOS / SMBv1 Exposure

netbios_smbv1_exposure

Inputs: Zeek: conn.log, smb.log, smb_mapping.log, smb_files.log, smb_cmd.log. Context/policy: segment/zone definitions, allowed host pairs, allowed segment pairs, netbios smbv1 policy.

Zeek evidence: **conn.log** (endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags); **smb.log** (SMB dialect/signing/session data); **smb_mapping.log** (share mapping names); plus smb_files.log, smb_cmd.log.

Detection: Identifies NetBIOS traffic on ports 137-139 or explicitly evidenced SMBv1 traffic crossing configured network-segment boundaries or involving public Internet addresses.

38. New Service Emergence in OT (Baseline Drift)

new_service_emergence_ot

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, service baseline policy, ignored hosts.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies services newly observed on configured OT hosts after an established baseline period, using Zeek service identification when available and protocol/port identity as a fallback.

40. NTP Sourced from Internet (Multi-Dest / OT)

ntp_internet_multi_dest_ot

Inputs: Zeek: ntp.log. Context/policy: segment/zone definitions.

Zeek evidence: **ntp.log** and supplies client/server NTP exchanges and timestamps.

Detection: Detects OT hosts using multiple distinct globally routable NTP servers instead of a centralized trusted time source, with optional trusted-server and external-server allow-list policy.

Detection modules 41-48

41. NTP Source Drift

ntp_source_drift

Inputs: Zeek: ntp.log. Context/policy: trusted NTP servers, ntp source drift policy.

Zeek evidence: **ntp.log** and supplies client/server NTP exchanges and timestamps.

Detection: Detects NTP clients that use servers outside explicitly trusted time infrastructure or change server after a capture baseline.

43. OT Host Using External / Internet DNS Resolver

ot_external_dns_resolver

Inputs: Zeek: dns.log. Context/policy: segment/zone definitions, asset inventory, explicit OT hosts.

Zeek evidence: **dns.log** and supplies client/resolver, query name/type, reply code and timestamp.

Detection: Detects OT/control-system hosts sending DNS queries to public Internet resolvers or resolvers outside the configured OT boundary instead of approved internal DNS infrastructure.

45. OT Outbound Internet Connection (Any Protocol)

ot_outbound_internet_any_protocol

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, asset inventory, allowed host pairs, approved external destinations.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Generic egress catch-all for sessions initiated by configured OT/control-system assets toward globally routable Internet destinations, irrespective of application protocol.

47. OT Protocol Role Reversal

ot_protocol_role_reversal

Inputs: Zeek: conn.log. Context/policy: ot role reversal policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects hosts that are established as OT protocol responders during the baseline and later begin originating that protocol toward peers.

42. OT Asset Gone Silent (Loss of Expected Periodic Comms)

ot_asset_gone_silent

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, asset inventory, allowed host pairs, explicit OT hosts.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Learns regular OT host/conversation cadence during a baseline period and identifies expected periodic communications that cease for materially longer than their established interval.

44. Certificate Long Validity / Self-Issued in OT Mgmt

ot_management_certificate_risk

Inputs: Zeek: ssl.log, x509.log. Context/policy: segment/zone definitions, management ports, ot certificate policy, certificate management policy.

Zeek evidence: **ssl.log** (TLS endpoints, SNI, version/cipher, fingerprint/JA3 data and certificate references); **x509.log** (subject/issuer/SAN, validity dates, key size and signature algorithm).

Detection: Identifies long-validity and self-issued TLS certificates presented by configured OT management endpoints.

46. OT Protocol Exposure

ot_protocol_exposure

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies industrial control protocols observed outside configured control-system network segments, including Modbus, DNP3, EtherNet/IP, BACnet, IEC 61850 MMS, and S7comm.

48. PLC Program / Logic Download or Firmware Update

plc_program_logic_firmware_update

Inputs: Zeek: modbus.log, s7comm.log, enip.log, dnp3.log. Context/policy: allowed host pairs, allowed/ignored hosts.

Zeek evidence: **modbus.log** (client/server, function/operation, exception/status and time); **s7comm.log** (client/PLC, function/operation and S7 error/result); **enip.log** (originator/target, CIP service/status and Forward_Open activity); plus dnp3.log.

Detection: Detects explicit ICS protocol operations associated with PLC program/logic transfer, firmware update, or DNP3 restart/file-transfer activity.

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49. PLC / RTU Peer Change

plc_rtu_peer_change

Inputs: Zeek: conn.log. Context/policy: asset inventory, allowed host pairs, controller hosts, plc rtu peer change policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects a new communication peer involving an authoritative PLC, RTU, controller, or IED after the capture baseline.

51. Port / Host Scanning

port_host_scanning

Inputs: Zeek: conn.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies rapid connection attempts from a single source to many ports on one host (port scan) or to many hosts on the same service/port (host sweep).

53. Public-to-Public Traffic in Capture

public_to_public_traffic

Inputs: Zeek: conn.log. Context/policy: allowed host pairs, public to public policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies traffic between two globally routable IP addresses when the monitored capture is explicitly expected to contain only internal ICS/OT communications.

55. Unexpected RDP with NLA Disabled

rdp_nla_disabled

Inputs: Zeek: rdp.log, conn.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **rdp.log** (RDP negotiation and NLA/CredSSP indicators); **conn.log** (endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags).

Detection: Identifies RDP sessions where protocol telemetry explicitly shows Network Level Authentication (NLA/CredSSP) was not used, plus lower-severity cases where NLA is explicitly not required.

50. Polling Cadence Disruption

polling_cadence_disruption

Inputs: Zeek: conn.log. Context/policy: polling cadence disruption policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects established periodic OT polling whose post-baseline interval shifts materially from its learned cadence.

52. Protocol on Unexpected / High-Risk Port

protocol_unexpected_high_risk_port

Inputs: Zeek: conn.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies Zeek-recognized application protocols running on unexpected ports and traffic using ports with a strong historical association with backdoors, reverse shells, proxies, or command-and-control tooling.

54. QUIC in OT Segments

quic_ot_segments

Inputs: Zeek: quic.log, conn.log. Context/policy: segment/zone definitions, quic ot policy.

Zeek evidence: **quic.log** (explicit QUIC/HTTP3 protocol evidence); **conn.log** (endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags).

Detection: Identifies QUIC/HTTP/3 traffic involving configured OT/control segments, with explicit protocol evidence preferred over lower-confidence UDP/443 heuristics.

56. Remote Access Session Anomaly

remote_access_session_anomaly

Inputs: Zeek: conn.log. Context/policy: remote-access ports, authorized remote-access sources, remote access session anomaly policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects authorized remote-access sources that fan out to unusual OT targets after a baseline.

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57. VNC / Remote Access Tool Exposure

remote_access_tool_exposure

Inputs: Zeek: vnc.log, conn.log. Context/policy: segment/zone definitions, allowed host pairs, remote access tool policy, allowed/ignored hosts.

Zeek evidence: **vnc.log** (VNC session evidence); **conn.log** (endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags).

Detection: Identifies VNC, pcAnywhere, TeamViewer, AnyDesk, Radmin, and configured remote-access tools, with higher priority for traffic involving OT/ICS segments or public Internet peers.

59. S7comm Unauthorized Write/Stop PLC

s7comm_unauthorized_write_stop

Inputs: Zeek: s7comm.log. Context/policy: authorized paths, s7comm control policy, approved function codes.

Zeek evidence: **s7comm.log** and supplies client/PLC, function/operation and S7 error/result.

Detection: Detects explicit Siemens S7comm write/download/control operations and PLC STOP commands, with optional policy checks for approved source-to-destination control paths.

61. SMB Admin-Share Access (IPC\$/ADMIN\$)

smb_admin_share_access

Inputs: Zeek: smb_mapping.log, smb.log, smb_cmd.log, conn.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **smb_mapping.log** (share mapping names); **smb.log** (SMB dialect/signing/session data); **smb_cmd.log** (SMB command activity); plus conn.log.

Detection: Identifies explicit SMB access to IPC\$ or ADMIN\$ and highlights repeated or multi-target patterns that can be consistent with remote administration or lateral movement.

63. SNMP Write / Set to OT Devices

snmp_write_ot_devices

Inputs: Zeek: snmp.log. Context/policy: segment/zone definitions, asset inventory, allowed host pairs, snmp write policy.

Zeek evidence: **snmp.log** and supplies manager/agent, PDU type, community/version and operation.

Detection: Detects SNMP SET operations or use of explicitly configured writable SNMPv1/v2c communities against OT devices, controllers, or managed network infrastructure beyond ordinary read-only polling.

58. Rogue DHCP Server / DHCP Activity on Static OT Segment

rogue_dhcp_static_ot

Inputs: Zeek: dhcp.log. Context/policy: segment/zone definitions, approved DHCP servers, static-address OT segments, dhcp ot policy.

Zeek evidence: **dhcp.log** and supplies client/server, DHCP message type, offer/lease address and time.

Detection: Detects DHCP offers or lease activity from unapproved servers, competing offers from multiple DHCP servers, and DHCP request/offer/lease activity on explicitly static or DHCP-prohibited OT segments.

60. Service Disappearance / Replacement

service_disappearance_replacement

Inputs: Zeek: conn.log. Context/policy: asset inventory, explicit OT hosts, service disappearance replacement policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects OT endpoints where a baseline service disappears and a different service appears after the baseline.

62. SMB Signing Disabled or NTLMv1 Detected

smb_signing_disabled_ntlmv1

Inputs: Zeek: smb.log, ntlm.log, conn.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **smb.log** (SMB dialect/signing/session data); **ntlm.log** (NTLM version/status and linked session evidence); **conn.log** (endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags).

Detection: Identifies SMB sessions explicitly observed without message signing or with signing disabled/not required, and NTLM authentication records that explicitly identify NTLMv1.

64. SOCKS / Open Proxy Behavior

socks_open_proxy_behavior

Inputs: Zeek: socks.log, http.log, conn.log. Context/policy: proxy behavior policy.

Zeek evidence: **socks.log** (SOCKS version/request and proxy endpoints); **http.log** (method, host/URI, User-Agent, status, body length and flow UID); **conn.log** (endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags).

Detection: Identifies hosts acting as SOCKS or HTTP CONNECT proxy endpoints using explicit protocol telemetry, without inferring proxy behavior from common ports alone.

Detection modules 65-72

65. TCP Reset / Connection-Abort Surge

tcp_reset_abort_surge

Inputs: Zeek: conn.log. Context/policy: tcp reset abort policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects bursts of reset/rejected/aborted TCP sessions against the same destination.

66. TLS Certificate Anomalies

tls_certificate_anomalies

Inputs: Zeek: ssl.log, x509.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **ssl.log** (TLS endpoints, SNI, version/cipher, fingerprint/JA3 data and certificate references); **x509.log** (subject/issuer/SAN, validity dates, key size and signature algorithm).

Detection: Identifies self-signed, expired, hostname-mismatched, weak-key, and unknown/untrusted-CA TLS certificates using Zeek ssl.log and x509.log when available.

67. TLS SNI Mismatch / Certificate Reuse Across Hosts

tls_sni_certificate_reuse

Inputs: Zeek: ssl.log, x509.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: **ssl.log** (TLS endpoints, SNI, version/cipher, fingerprint/JA3 data and certificate references); **x509.log** (subject/issuer/SAN, validity dates, key size and signature algorithm).

Detection: Detects TLS sessions whose SNI/server identity does not match the presented certificate and certificates reused broadly across unrelated server hosts/domains.

68. Unexpected DHCP Server

unexpected_dhcp_server

Inputs: Zeek: dhcp.log. Context/policy: approved DHCP servers, unexpected dhcp server policy.

Zeek evidence: **dhcp.log** and supplies client/server, DHCP message type, offer/lease address and time.

Detection: Detects DHCP server responses from sources outside the explicitly configured trusted DHCP infrastructure.

69. Unexpected Multicast Behavior

unexpected_multicast_behavior

Inputs: Zeek: conn.log. Context/policy: asset inventory, allowed multicast groups, unexpected multicast policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects OT hosts initiating multicast traffic to groups outside explicitly allowed multicast destinations.

70. Unknown / Rogue Devices

unknown_rogue_devices

Inputs: Zeek: conn.log, arp.log, dhcp.log, dns.log. Context/policy: asset inventory.

Zeek evidence: **conn.log** (endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags); **arp.log** (sender/target IP-MAC pairs, opcode and timestamp); **dhcp.log** (client/server, DHCP message type, offer/lease address and time); plus dns.log.

Detection: Identifies locally observed IP or MAC addresses that are not present in the supplied asset inventory.

71. Unusual Outbound Data Volume

unusual_outbound_data_volume

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, outbound volume policy, data volume policy.

Zeek evidence: **conn.log** and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Identifies connections with unusually large Zeek orig_bytes leaving local networks, especially to external destinations or across explicitly configured trust boundaries.

72. UPnP / SSDP Exposure & IGD Port Mapping

upnp_ssdp_igd_port_mapping

Inputs: Zeek: upnp.log, upnp_igd.log, ssdp.log, conn.log. Context/policy: segment/zone definitions.

Zeek evidence: **upnp.log** (UPnP request/response evidence); **upnp_igd.log** (IGD mapping operations); **ssdp.log** (SSDP discovery/announcement evidence); plus conn.log.

Detection: Detects explicit UPnP Internet Gateway Device port-mapping activity and unexpected SSDP exposure while avoiding port-only inference from UDP/1900.

Detection modules 73-75

73. VLAN Tag Mismatch / Double-Tag Indicators

vlan_tag_mismatch_double_tag

Inputs: Zeek: conn.log. Context/policy: segment/zone definitions, vlan policy, allowed outer/inner VLAN pairs, allowed VLAN IDs.

Zeek evidence: conn.log and supplies endpoints, ports/service, timestamps, bytes/packets, connection state and VLAN tags.

Detection: Detects explicit 802.1Q VLAN IDs in Zeek conn.log that fall outside configured VLAN policy, and double-tagged/Q-in-Q connections that may indicate VLAN hopping or unexpected trunk/provider tagging.

75. Weird.log Protocol Violations

weird_protocol_violations

Inputs: Zeek: weird.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: weird.log and supplies weird name/reason, protocol/endpoints and timestamp.

Detection: Analyzes Zeek weird.log for malformed packets, checksum anomalies, unexpected protocol states, and other protocol-specification violations.

74. Weak or Broken TLS/SSL

weak_broken_tls_ssl

Inputs: Zeek: ssl.log, x509.log. Context/policy: module defaults / capture evidence only.

Zeek evidence: ssl.log (TLS endpoints, SNI, version/cipher, fingerprint/JA3 data and certificate references); x509.log (subject/issuer/SAN, validity dates, key size and signature algorithm).

Detection: Identifies deprecated SSL/TLS versions, weak cipher suites, and weak certificate signature algorithms when the relevant Zeek logs are present.

Source baseline: eleVADR Dataset 18 real-site validation checkpoint; 75 registered modules. This reference summarizes the detector source code and module metadata in that checkpoint.